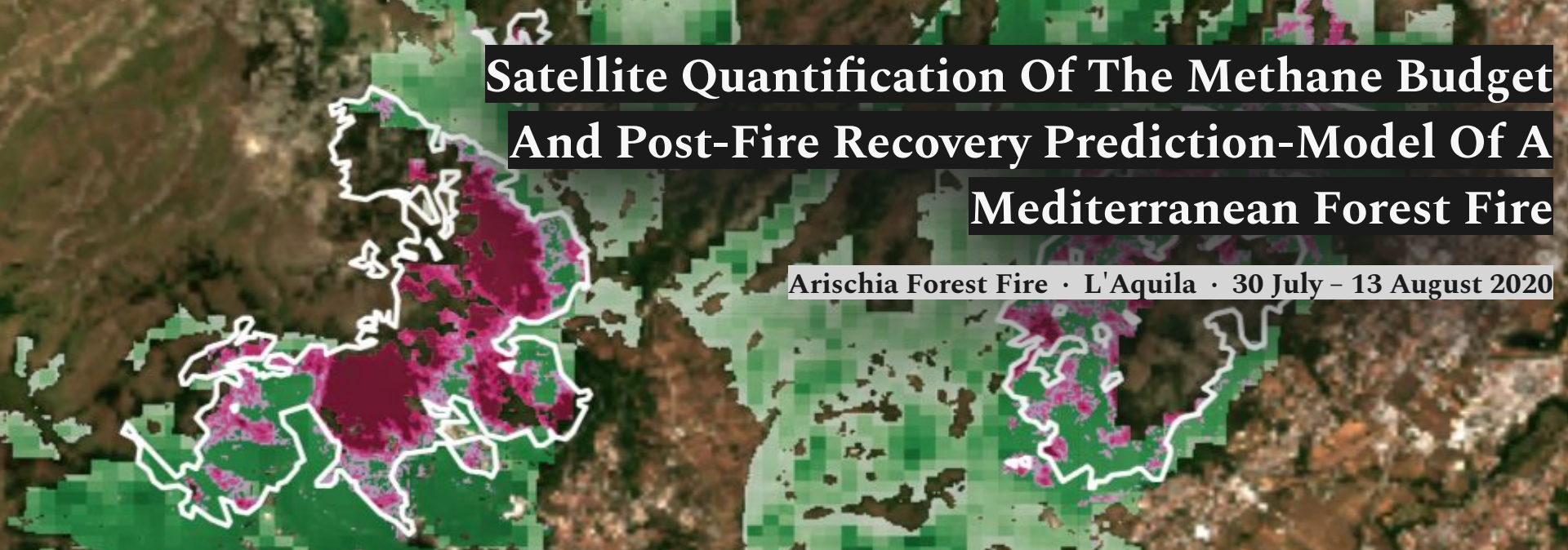




Satellite Quantification Of The Methane Budget And Post-Fire Recovery Prediction-Model Of A Mediterranean Forest Fire

Arischia Forest Fire · L'Aquila · 30 July – 13 August 2020



SAPIENZA
UNIVERSITÀ DI ROMA

PROFESSOR

Ferdinando Nunziata

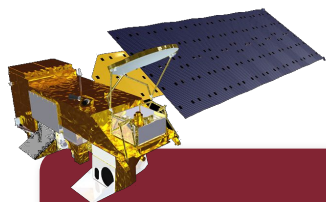
EARTH OBSERVATION PROJECT

Vinoth Emberumal

Matricola 2248220

MSc Space & Astronautical Engineering

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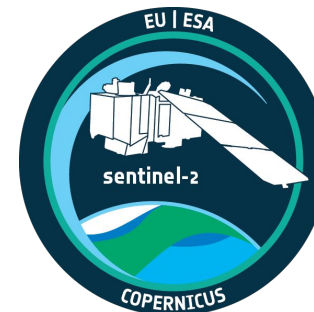
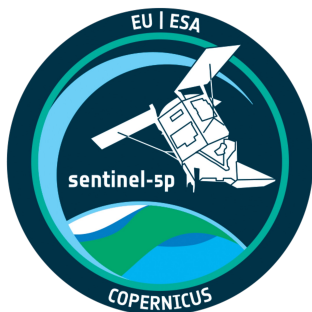
cross-method · vs reference

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Why methane, and not carbon dioxide?

24.2 kt

CO₂ released

65.07 t

CH₄ released

0.25 %

CH₄ share of mass

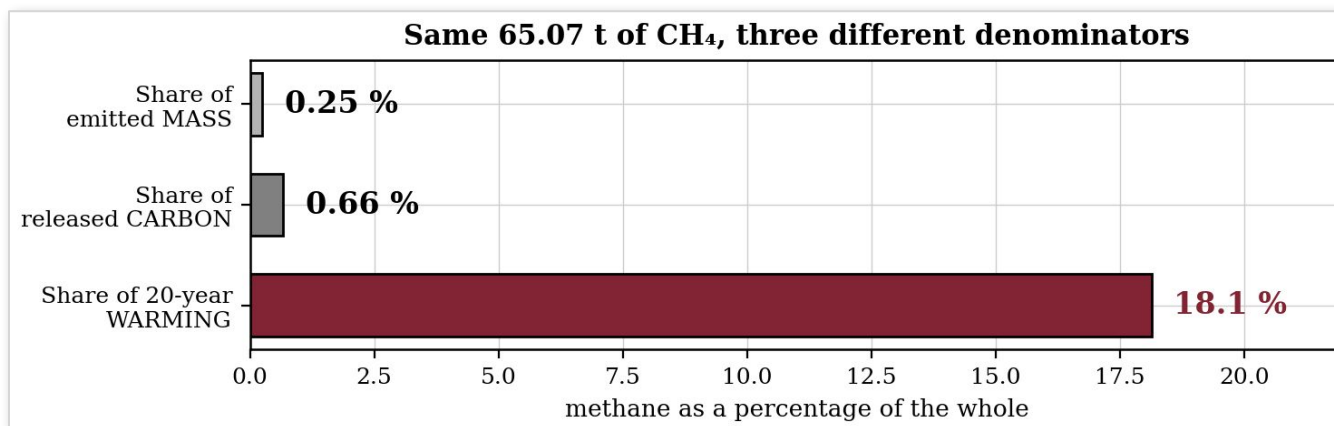
82.5 ×

heat trapped per kg to eCO₂

12 yr

atmospheric lifetime

- **1 kg of CH₄** traps the heat of **82.5 kg of CO₂** over the first twenty years.
- A **powerful thermal accelerator**: it sustains that forcing for **12–20 years**, then degrades into CO₂.
- So **0.25 %** of the emitted mass carries **18.1 %** of the twenty-year warming.



1.5 – 4.5

kg ha⁻¹ yr⁻¹ oxidised

309 ha

sink destroyed

14 days

of burning

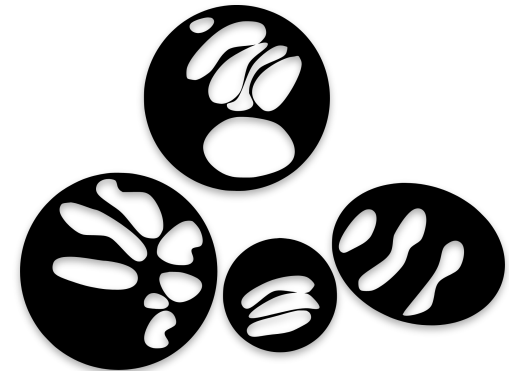
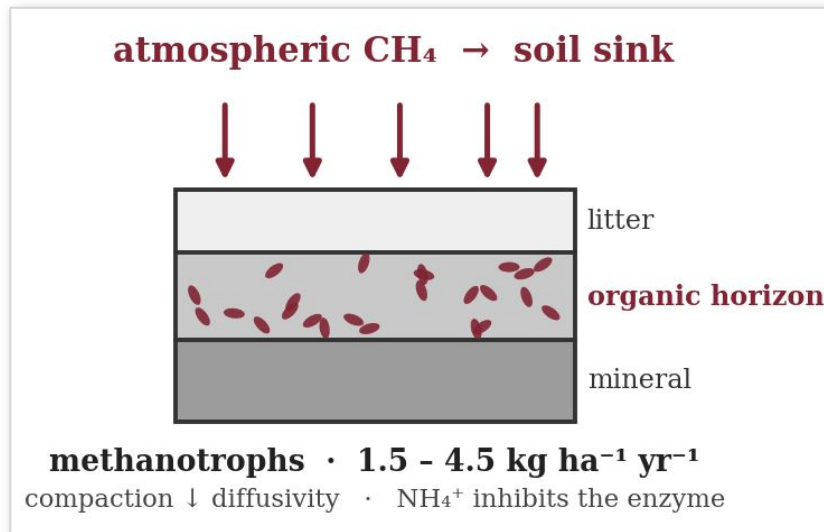
0.46–1.39 t

yearly removal lost

47–140 yr

the debt

- Forest soil is the **largest biological land sink** for methane — **methanotroph bacteria** oxidise 1.5–4.5 kg ha⁻¹ yr⁻¹.
- Fire **emits a pulse and destroys the sink** in the same event.
- The 14-day pulse equals **47–140 years** of this forest's own removal capacity.



Methylococcus
capsulatus

5

satellite inputs

5

processing steps

EPSG:3035

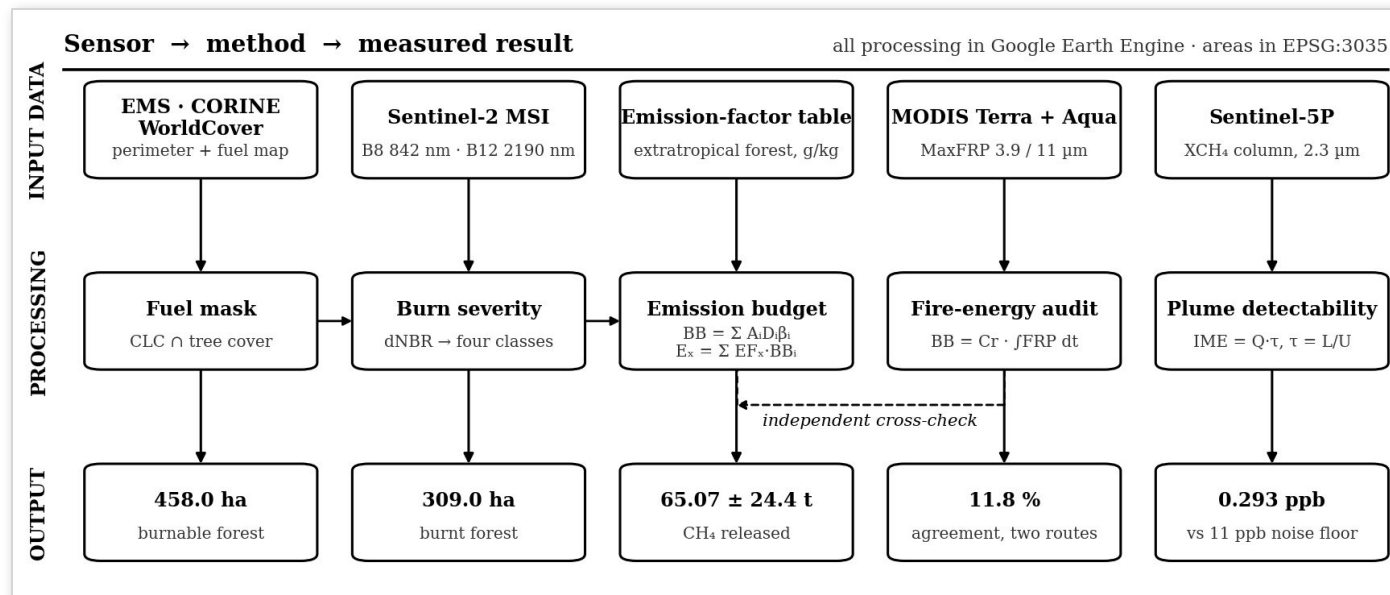
area projection

11 yr

archive depth

GEE

all server-side



- Fixed **15 Jun – 15 Sep** window every year · **SCL** cloud mask · **median** compositing.
- An **unburnt control forest** runs through every temporal step.

638.9 ha

EMS perimeter

458.0 ha

burnable forest

309.0 ha

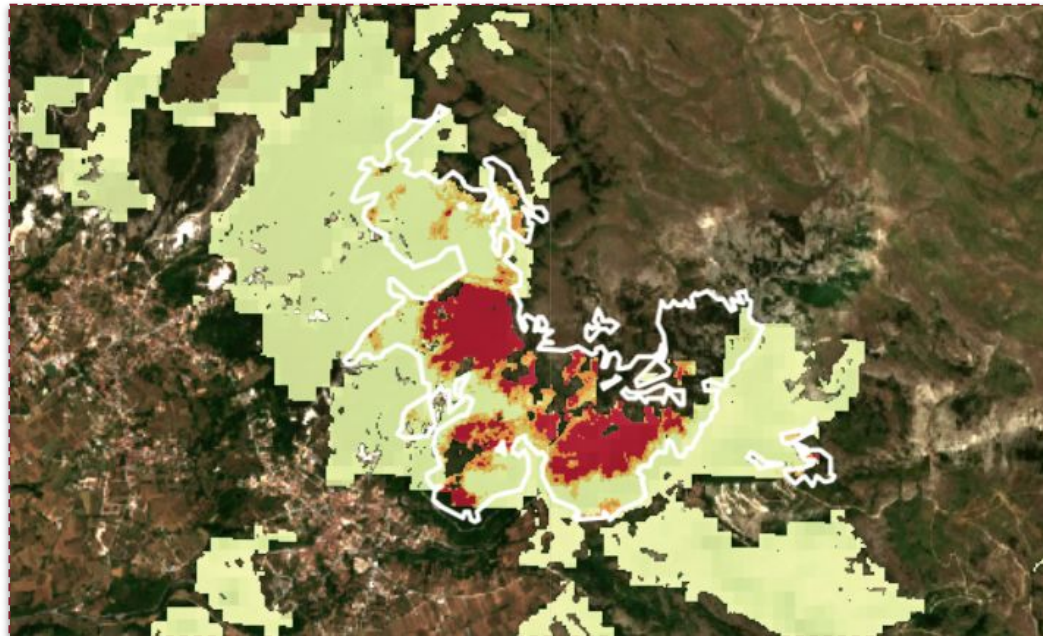
burnt forest

42.4° N

central Apennines

14 days

30 Jul – 13 Aug 2020



- White outline — **Copernicus EMS EMSR449** operational delineation, used as reference geometry.
- Red - orange — **burnt forest**; green — **surviving canopy** inside and around the perimeter.

28.3 %

of perimeter is non-fuel

67 %

of fuel actually burned

48.5 t/ha

biomass consumed

15.0 kt

total biomass

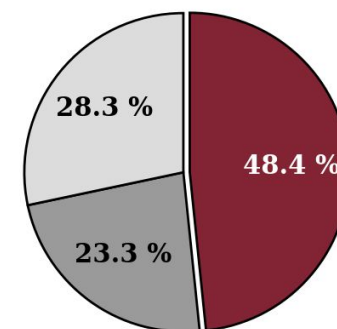
59.6 t

published CH₄ anchor

Layer	Value	Source
Mapped fire perimeter	638.9 ha	Copernicus EMS · EMSR449
Non-combustible ground	180.9 ha · 28.3 %	CORINE ∩ WorldCover
Burnable forest	458.0 ha	fuel mask
Burnt forest, dNBR ≥ 0.16	309.0 ha	Sentinel-2 severity map
Unburnt forest inside perimeter	149.0 ha	fuel mask – burnt

Level-4 thematic products derived from satellite data

638.9 ha mapped perimeter



■ Burnt forest 309.0 ha
■ Unburnt forest inside perimeter 149.0 ha
■ Non-fuel removed 180.9 ha

311

broadleaf

312

coniferous

313

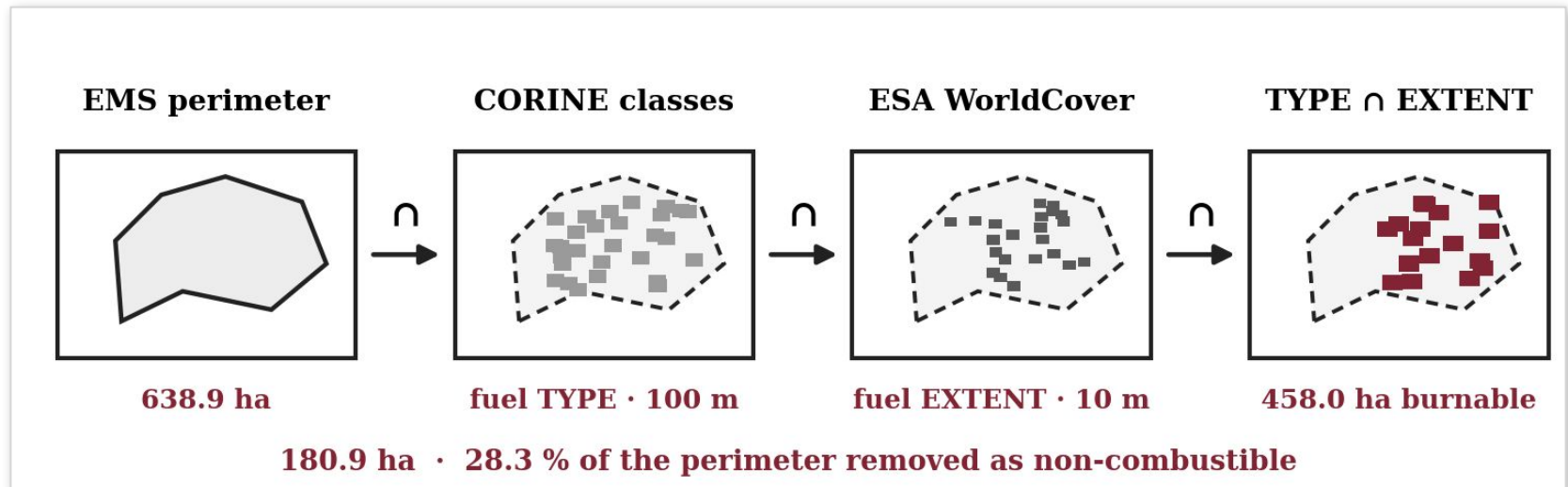
mixed forest

324

transitional woodland

10 m

WorldCover extent



- CORINE gives the **class**; WorldCover 2020 gives the **extent** at the fire's own epoch.
- The intersection removes **180.9 ha — 28.3 %** of the perimeter, with **no tuned parameter**.

0.16

dNBR threshold

4

severity classes

309 ha

burnt forest

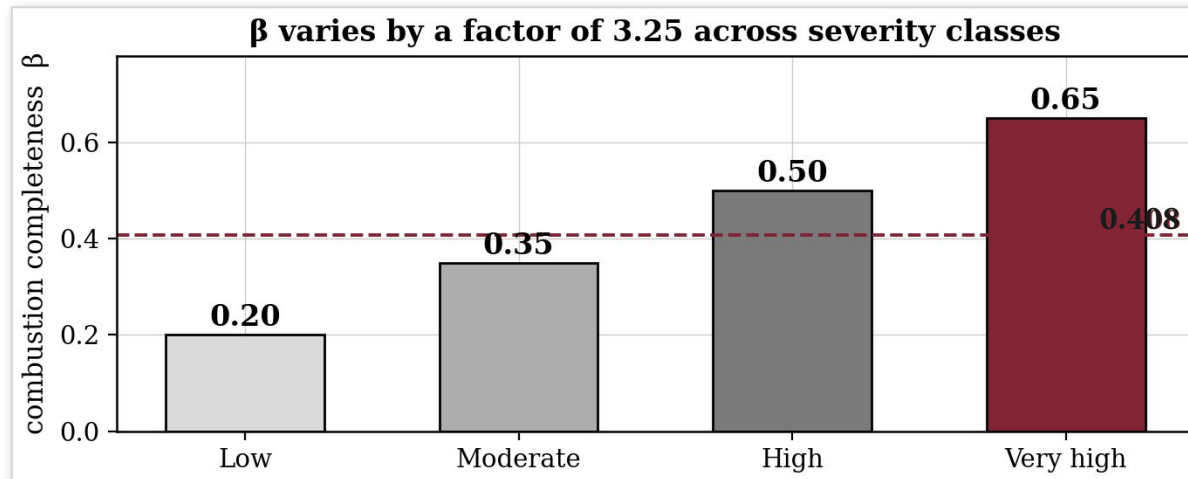
3.25 ×

β range across classes

0.408

β effective

- $NBR = (B8 - B12)/(B8 + B12)$ · $dNBR = NBR_{pre} - NBR_{post}$
- β is the fraction of available fuel actually consumed — it is **why severity must be mapped**, not averaged.



200,000

draws

± 40 %

EF, the dominant term

± 25 %

β

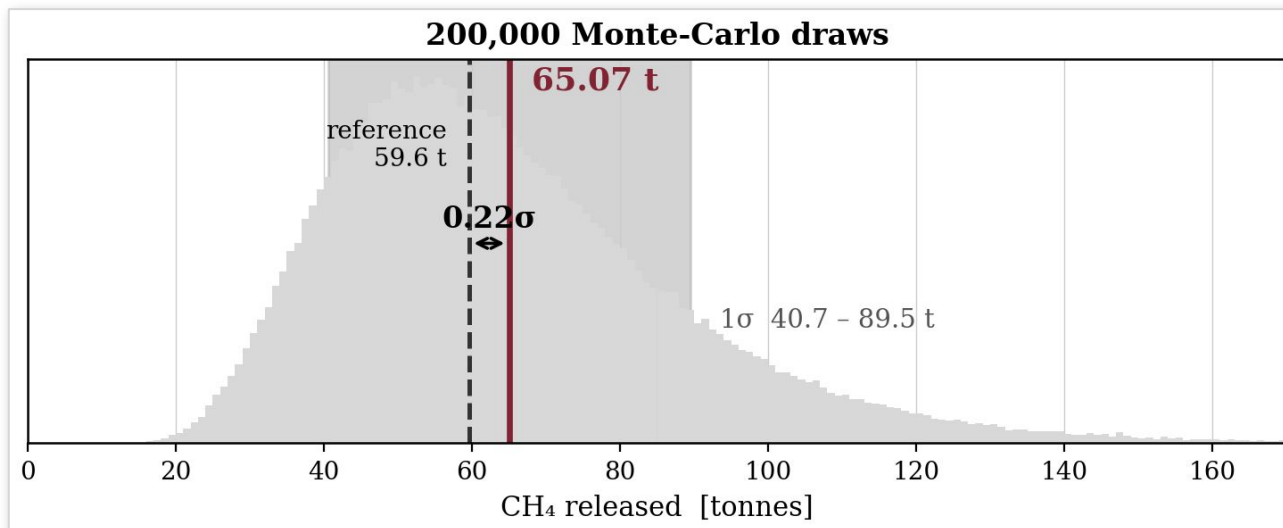
± 20 %

fuel load D

± 5 %

burnt area A

- $E = EF \times A \times D \times \beta$ — the terms **multiply**, so relative variances add and the result is **right-skewed**.
- $(\sigma_E / E)^2 \approx \sum (\sigma_i / x_i)^2$ — only a simulation gets the **shape** right, not just the width.



Instruments and what each one yields

5 · QUANTIFICATION

10 m

Sentinel-2 GSD

5 day

S-2 revisit

1 km

MODIS fire pixel

4 / day

MODIS overpasses

11 ppb

TROPOMI 1σ

Instrument	Measurement	Method	Result
Sentinel-2 MSI	reflected solar, B8 & B12	$NBR \cdot dNBR \cdot RBR$	309.0 ha , four severity classes
MODIS Terra + Aqua	mid-IR and thermal-IR radiance	$FRE = \int FRP dt \cdot BB = C_r \cdot FRE$	16,751 t biomass, independent route
Sentinel-5P TROPOMI	SWIR absorption at $2.3 \mu m$	$\tau = L/U \cdot IME = Q \cdot \tau \cdot \Delta XCH_4$	0.293 ppb against an 11 ppb floor

Three sensors, three corners of the resolution triangle — **spatial, temporal, spectral.**

The quantified budget

≈ 15.0 kt
biomass consumed

24.2 kt
CO₂

1,416 t
CO

167 t
PM_{2.5}

65.07 t
CH₄

Species	Mass [t]	t ha ⁻¹	% of mass	% of carbon	EF [g kg ⁻¹]
CO ₂	24,244.07	78.46	93.63	89.74	1,618
CO	1,416.45	4.58	5.47	8.24	94.5
PM _{2.5}	167.45	0.542	0.65	1.36	11.2
CH ₄	65.07 ± 24.4	0.211	0.25	0.66	4.34

Seiler & Crutzen (1980): $BB = \sum_i A_i \cdot D \cdot \beta_i$ · implied EF_CH₄ **4.34 g/kg** · carbon balance closes at **104.6 %**

Where the carbon actually went

5 · QUANTIFICATION

93.6 %

CO₂ share of mass

0.25 %

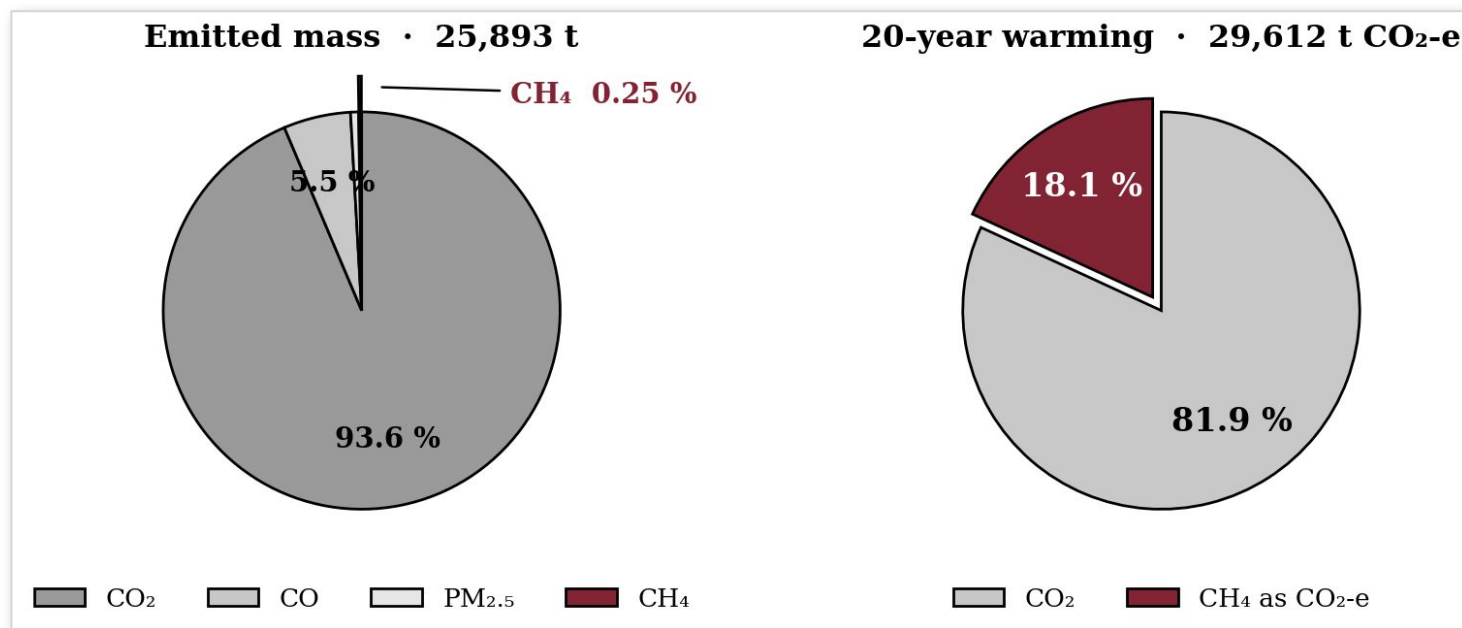
CH₄ share of mass

65.05t CH₄ = 5368t Co2

Methane and Co2 Equivalent

24.2 kt to 29.6 kt

CO₂-e over 20 yr



- The same 65.07 t changes its weight entirely depending on the **denominator** you choose.
- A CO₂-only inventory would report the left pie and stop — and miss **one fifth** of the near-term impact.

11.8 %

optical vs thermal

104.6 %

carbon closure

0.22 σ

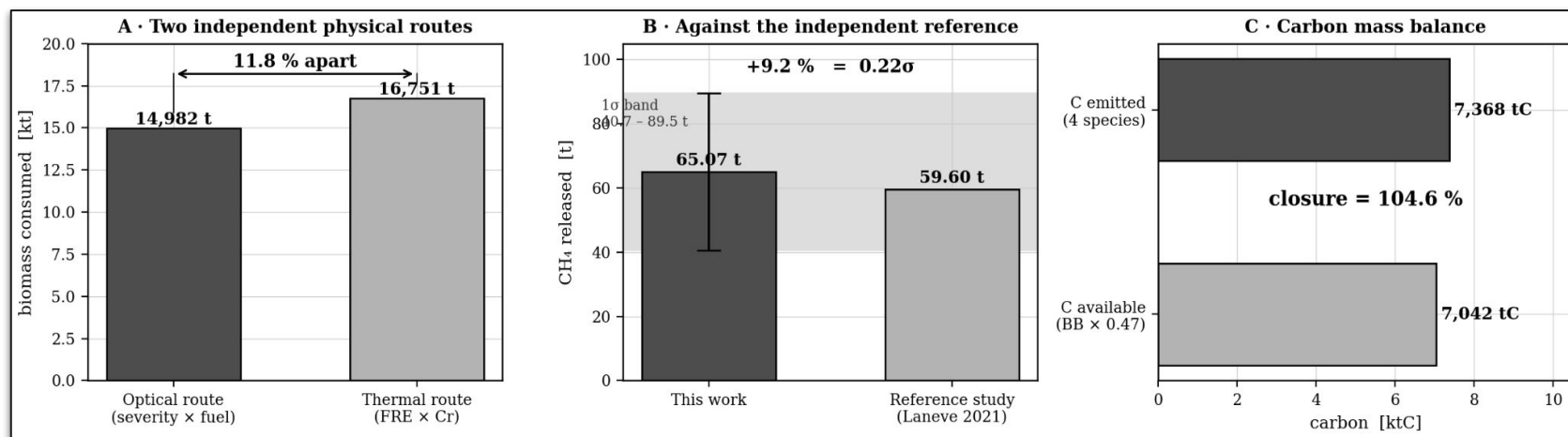
from the reference

-1.22 %

control drift, 11 yr

14.5 %

diurnal duty cycle



- Two physically unrelated observables — **reflected solar** (**Optical**) and **emitted mid-infrared** (**Thermal**) — converge on the same fuel mass. $BB = Cr \cdot FRE$, $Cr = 0.368 \text{ kg MJ}^{-1}$ → $FRE = 16,751 \text{ t}$ (Diurnal Duty Cycle)
- The carbon balance closes at **104.6 %** with emission factors taken from the literature and **never tuned**.

+16.4 %

on CO₂ emission factor

+9.2 %

on CH₄

1,390

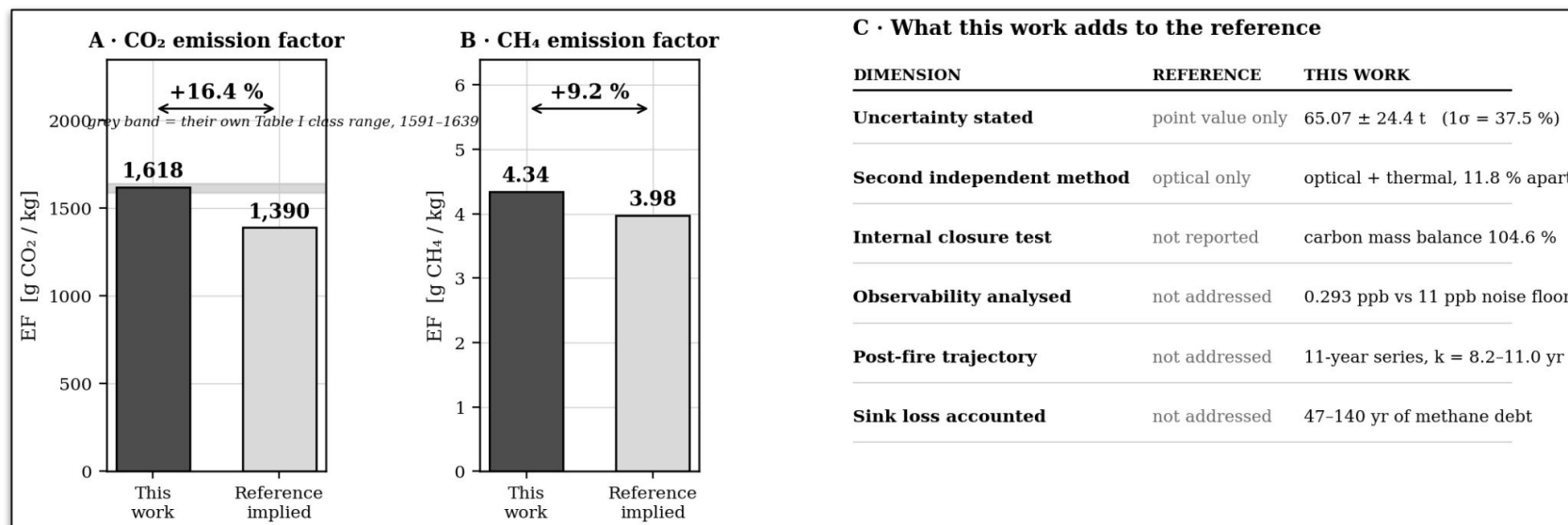
g/kg implied by them

1,591–1,639

their own Table I range

1.164 ×

predicted, and obtained



- The corresponding methane comparison gives **4.34 g kg⁻¹** against an implied **3.98 g kg⁻¹**
- I predicted my CO₂ would be **1.164× higher** before computing it. It was, **exactly**.

The Interactive dashboard

7 · TOOLS

7

map layers

GeoTIFF

raster export

GeoJSON

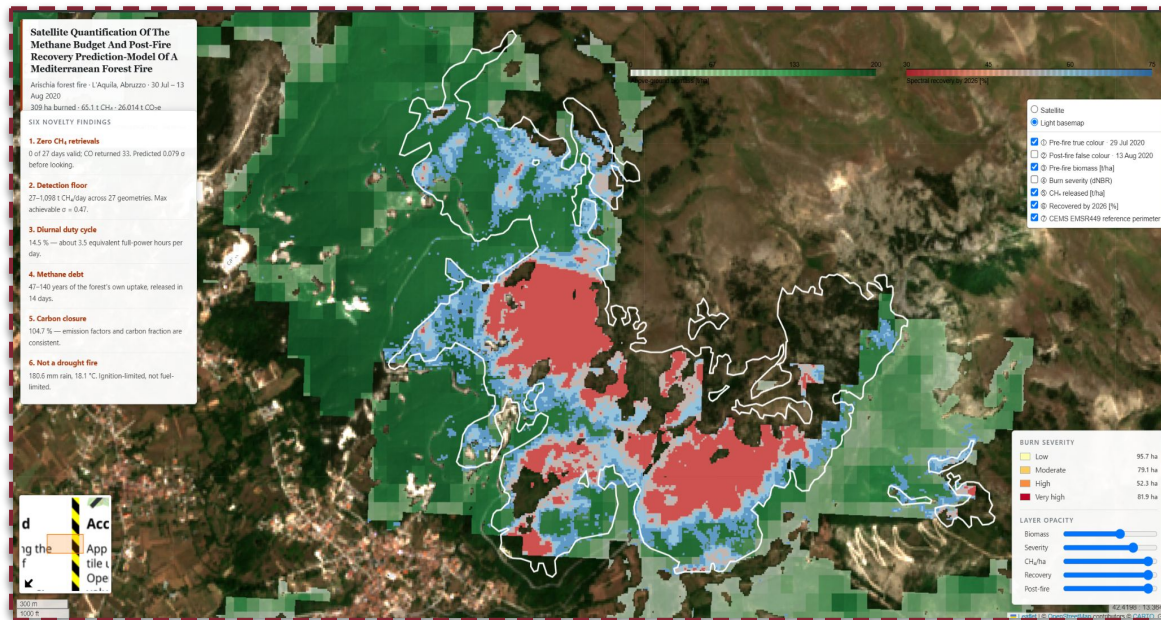
vector export

folium

Leaflet front-end

1 file

fully offline



Layer	What it shows
① pre-fire true colour	the forest the day before
② post-fire false colour	the scar, before any index
③ fuel map	what could burn — 458.0 ha standing biomass = 36,740 t
④ burn severity	309.0 ha, four classes
⑤ CH ₄ per hectare	0.211 t ha ⁻¹ mean - β Spread
⑥ recovery by 2026	NBR(2026) ÷ NBR(2016-19 baseline) — a spectral proxy for canopy : forest health
⑦ EMS perimeter	reference truth

The Processed Satellite data Mapped in to a **HTML DASHBOARD**

Google Colab

browser only + Python 3.14

16

code cells

Earth Engine

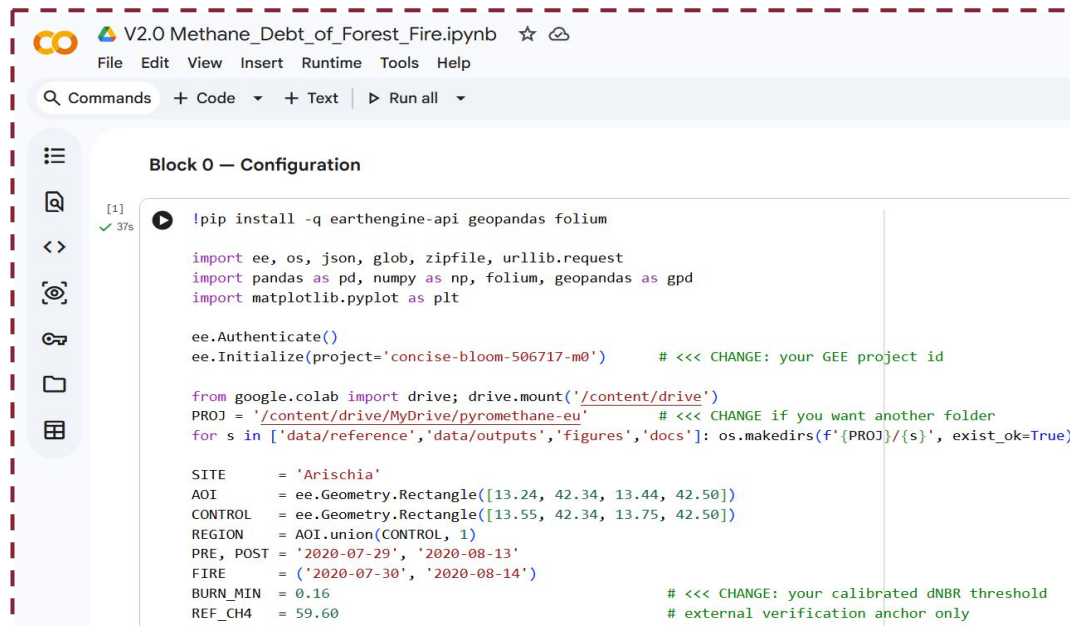
server-side

MIT

open licence

1 register

every number logged



```
[1] ✓ 37s
!pip install -q earthengine-api geopandas folium

import ee, os, json, glob, zipfile, urllib.request
import pandas as pd, numpy as np, folium, geopandas as gpd
import matplotlib.pyplot as plt

ee.Authenticate()
ee.Initialize(project='concise-bloom-506717-m0') # <<< CHANGE: your GEE project id

from google.colab import drive; drive.mount('/content/drive')
PROJ = '/content/drive/MyDrive/pyromethane-eu' # <<< CHANGE if you want another folder
for s in ['data/reference', 'data/outputs', 'figures', 'docs']: os.makedirs(f'{PROJ}/{s}', exist_ok=True)

SITE = 'Arischia'
AOI = ee.Geometry.Rectangle([13.24, 42.34, 13.44, 42.50])
CONTROL = ee.Geometry.Rectangle([13.55, 42.34, 13.75, 42.50])
REGION = AOI.union(CONTROL, 1)
PRE, POST = '2020-07-29', '2020-08-13'
FIRE = ('2020-07-30', '2020-08-14')
BURN_MIN = 0.16 # <<< CHANGE: your calibrated dNBR threshold
REF_CH4 = 59.60 # external verification anchor only
```

— Everything runs in the **browser** — no local software, no downloads.

— **Earth Engine** evaluates server-side: eleven years of Sentinel-2 in seconds rather than two terabytes.

— Every computed value is written to **one results register**, printed as a single table.

— Repository carries the **licence, environment file and exports** — the work is reproducible, not just presentable.

11 yr

Sentinel-2 composites

8.2 – 11.0 yr

time constant k

36 – 67 %

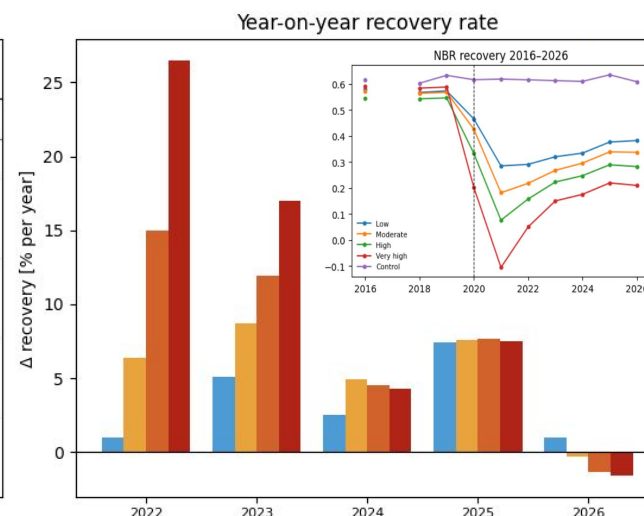
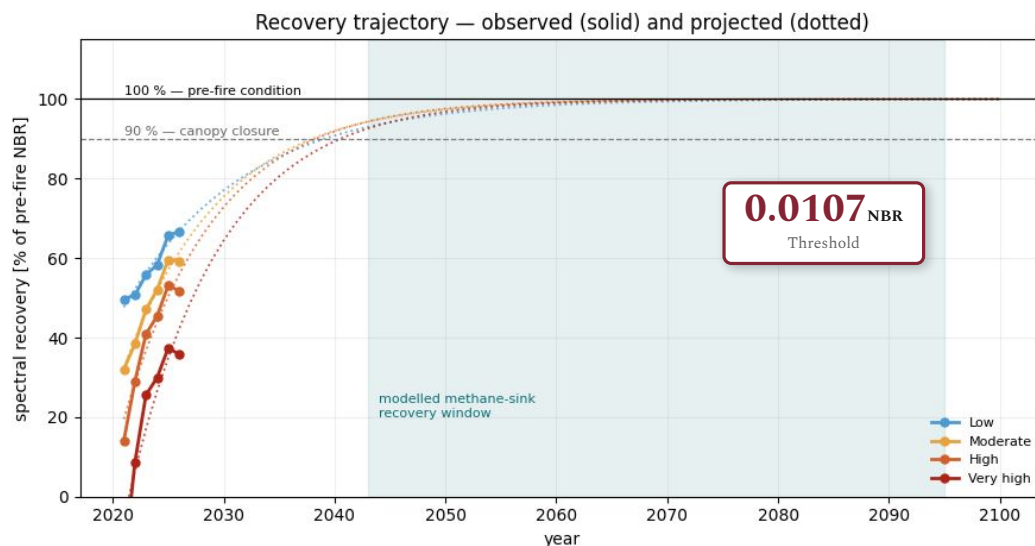
recovered by 2026

2038 – 2040

90 % canopy

-1.22 %

control drift



Severity	2026	k [yr]	90 %	Canopy Recovery 100 %
Low	66.6 %	11.0	2039	2057
Moderate	59.4 %	8.9	2038	2053
High	52.0 %	8.3	2038	2052
Very high	36.0 %	8.2	2040	2054

— $\log(\text{baseline} - \text{NBR})$ is linear in time
 $\Leftrightarrow$ recovery is **exponential** — the form is validated by the data.

— Severity sets **how deep the hole is**, not how fast the forest climbs out.

2038 – 2040
90 % canopy

2052 – 2058
full canopy

1.3 – 2.0 ×
microbial lag

2043 – 2095
CH₄ sink returns

≈ 50 yr
width of the window

Severity	Canopy 90 %	Canopy full	CH ₄ sink (modelled)
Low	2039	2058	2045 – 2095
Moderate	2038	2053	2043 – 2085
High	2038	2052	2044 – 2083
Very high	2040	2054	2047 – 2089

WHERE EARTH OBSERVATION STOPS

The methanotroph community lives in the **soil organic horizon**.

No optical, thermal or microwave band reaches it.

Canopy dates are **measured**; the sink column is **modelled**.

— Restoring the methane sink is **not the same problem** as restoring biomass — it depends additionally on the **soil microbial cycle** and the **megaherbivore cycle**.



Three proposals - European BIOMASS restoration policy

8 · RECOVERY & OUTLOOK

1

suppress at source -Zeolite

2

restore the sink

3

conserve the engineers

EU

Cohesion & Solidarity funds

0

EO Limitation: cover wildfire
CH₄

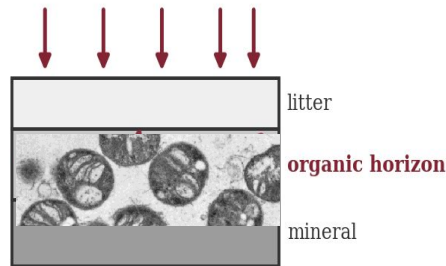
Cu-loaded zeolite bed

CH₄ in · smouldering front



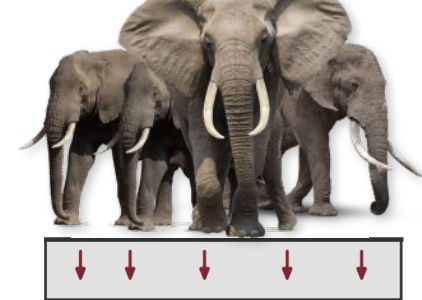
spent medium swapped out → asphalt / concrete

atmospheric CH₄ → soil sink



methanotrophs · 1.5 - 4.5 kg ha⁻¹ yr⁻¹
compaction ↓ diffusivity · NH₄⁺ inhibits the enzyme

soil structure · moisture · nutrient turnover



lower fuel continuity → lower fire SEVERITY

IMMEDIATE SUPPRESSION

- **Cu-loaded zeolite** beds oxidise concentrated CH₄ at the smouldering front.
- Product removed as a **solid** for industrial reuse.

RESTORE THE METHANE SINK

- **No ammonium fertiliser** — it inhibits the oxidising enzyme.
- **Microbial Cultivations** - Aerobic Condition

MEGAHERBIVORE CONSERVATION

- Large herbivores maintain **soil structure and nutrient turnover** - Maintain **C - Cycle**
- They lower fuel continuity — and therefore **severity**.

65.07 t

CH₄ released

±37.5 %

1-sigma

11.8 %

two routes agree

1- σ limit

from the reference

47–140 yr

methane debt

1. Quantification - I measured **65.07 tonnes** of methane from a **309 hectare** fire, and two physically independent satellite methods agree to **11.8 %**.

2. Validation - That methane is **0.25 %** of the emitted mass and **18.1 %** of the twenty-year warming — agreeing with the published study to **0.22 σ** .

3. Prediction Model - The forest will owe it back for **47 to 140 years** — and restoring that is a **soil-microbial** problem, not only a biomass one.

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Thankyou

PROFESSOR

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